

# Supplementary material

When does fusing hand-crafted knowledge with learned representations pay? A cost-normalized benchmark of stacking, substitution and interference

## S1. The self-supervised margin, at every measured fraction

The prior’s own gain envelopes are in the article (Tables 4, 5 and 6). This table gives the remaining envelope those tables summarize in prose: the margin of a  $2\times$ -compute SimCLR initialization over the  $1.02\times$  prior on CIFAR-100 with ResNet-18, under plain crop-and-flip augmentation, at every measured data fraction. Every entry is a paired difference against a shared baseline arm under the article’s frozen recipe, so a positive entry is a fraction where the  $2\times$ -compute initialization wins.

The shape is the point: the margin is unimodal. The self-supervised advantage is a mid-band phenomenon that starves below 1–2%, where too few images are available to contrast, and decays to nothing by 50%, where the data itself dominates and every intervention converges.

Table 1: SimCLR initialization ( $2\times$  compute) minus the prior ( $1.02\times$ ), CIFAR-100 / ResNet-18, plain augmentation. Three seeds per cell. At  $5\times$  pre-training compute the margin instead runs +3.1 to +9.2 over 1–15%, narrowing to +2.6 at 25% (article, Sec. 8).

| Frac. | SSL – prior | Frac. | SSL – prior |
|-------|-------------|-------|-------------|
| 1%    | +0.85       | 10%   | +5.01       |
| 2%    | +2.38       | 15%   | +4.09       |
| 3%    | +2.31       | 50%   | +0.73       |
| 5%    | +3.90       | 100%  | +0.22       |
| 7%    | +4.99       |       |             |

## S2. The law’s components

The article states the decomposition  $\Delta = G + \text{readout}$  and audits the sign of its second term. The component curves behind that audit are collected here.

**Feature-gain curves.**  $G$  is non-monotone in data and dataset-specific in shape: CIFAR-10 4.81, 5.52, 3.97, 0.65 at 0.5, 1, 2.5, 5k images; CIFAR-100 4.16, 5.14, 6.26, 3.55; Tiny-ImageNet falls monotonically 4.19, 2.66, 1.70 (1, 5, 10k); STL-10 4.70, 4.97, 3.20. Where the *readout* term is flat, the envelope peak coincides with the  $G$  peak: the envelope’s shape *is* the feature-gain curve.

**The *readout* branch.** Mapped on ceiling-free CIFAR-10: +1.56 at baseline 39.2, +1.14 at 51.7 and +0.44 at both 69.1 and 80.7, positive throughout and decaying toward sufficiency. Below the crossing it reaches  $-2.7$  at bases 8.9 and 5.3. The bracket [31.8, 40.3] used throughout the article is pinned by a Swin cell still negative at 31.8 and a convolutional cell already positive at 40.3.

**Label-space crossings.** On byte-identical CIFAR-100 pixels, a full  $2 \times 2$  of training against evaluation label space (100-way vs. 20-way) moves  $G$  by at most  $1\sigma$ . On Tiny-ImageNet the same crossing finds two real opposing effects (evaluation-space  $+1.9$ , checkpoint-set  $+1.6$ , both  $\sim 5\sigma$ ) that cancel on the diagonal, which is why  $G$  comparisons in the article always fix the evaluation space.

### S3. An illustrative panel, and why it is not evidence

The measurements behind the prior’s feature effect are the alignment statistic and the linear evaluation of the article’s Sec. 5.5. The panel below only illustrates them, and is reproduced precisely because it shows the effect *less* clearly than the numbers do.

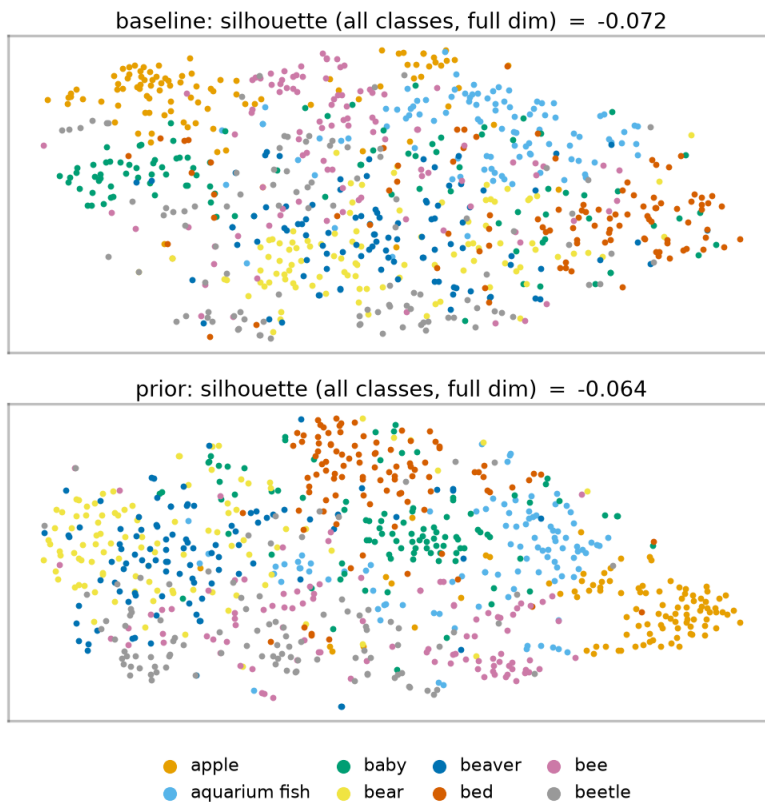


Figure 1: t-SNE of frozen penultimate test features, first eight classes, baseline (top) and prior arm (bottom), CIFAR-100 at 5%. The silhouette in each title is computed over *all* classes in the *full* feature space and moves only from  $-0.072$  to  $-0.064$ : this projection does *not* show the  $+6.3$ -point feature gap measured at this cell.

### S4. Per-domain notes

The article’s domain envelopes (its Table 5) are summarized there; these are the per-dataset readings behind them, including the one confound that changes how a column should be read.

EuroSAT: the prior beats SimCLR at 7 of 11 fractions ( $+0.6$ – $+1.9$  in the 1–10% band), consistent with SimCLR’s photographic view-invariances transferring poorly to satellite statistics while

the spectral target is domain-agnostic. DTD is the reverse, SimCLR dominating from a few hundred images, and Food-101 is SSL territory on convolutional backbones. PathMNIST declines with added data for *every* method above  $\sim 10\text{--}15\%$  (92.0 to 86.7 from 10% to 100%), the signature of its center-shifted test split, so only its low-data cells are interpretable; the article states this beside the table it affects. The domain axis does not split “photo vs. non-photo”: each domain’s statistics decide which currency is scarce.

## S5. Reproducibility

The article’s Data Availability statement names the repository and what it holds. This is the operational detail behind it.

A single entry point trains any cell from its configuration name and seed. Regenerating the result tables from the run records runs a small set of exporters, not one command, one per task family, covering classification, dense prediction and detection; we state that precisely because believing the one-command version once cost us stale tables. A further released script re-runs the sign-law audit with the seed-paired uncertainty the article’s Table 7 reports. Subset indices, filter-bank fingerprints and per-run environment records are released with the runs they describe. Training refuses to overwrite a completed cell, holds an exclusive per-run lock and writes checkpoints atomically (two of this study’s incident classes made these guards necessary), and mirrors are verified by *loading and evaluating* checkpoints against their recorded accuracies, not by file listing. The full harness is released at the repository cited on the article’s title page; an archival DOI will be minted on acceptance.

## S6. The dataset table, in words

Table 3 of the article lists every dataset with its training split size, class count and resolution as trained, and its caption carries the identity accounting. This is the description that accompanied it.

Six visual domains are represented, with the two ImageNet stages supplying the data-scale and the resolution-and-model-scale axes. Sources of variable native resolution are squash-resized once, offline, and the resized copies are what every arm consumes. Five relabeled controls derived from CIFAR-100 and Tiny-ImageNet share their parents’ pixels exactly and differ only in label space; they reuse their parent’s indices and share its row instead of appearing separately. Four further identities are band subsets of two parent instruments, sharing those instruments’ tiles and differing only in which channels each arm receives. Together these account for the gap between the twenty dataset identities counted in the article’s sign-law audit and the eleven independent image sources they are drawn from.

## S7. Selected controls

Three controls the article cites in passing. The first is the one a reader of the low-data gains needs, and the article states its result there as well.

**Contamination.** Re-evaluating the low-data CIFAR-100 cells on ciFAIR-100, which replaces the test set’s 927 near-duplicates of training images, moves the prior’s gain by at most 0.20 points at any fraction; both arms eat identical contamination, so it cancels in  $\Delta$ .

**Granularity, decisively.** Relabeling CIFAR-100’s fixed subsets with their 20 coarse labels (byte-identical pixels, identical steps) reproduces the full gain at matched data (+5.84 against +5.30 at 5%): gain follows data and task performance, not class count. The mirrored Tiny-ImageNet control (positional, visually incoherent coarse groups) is the cell that falsified our own earlier account: class count dropped tenfold, baseline accuracy did not rise, and no *readout* boost appeared, exactly as the article’s Eq. 1 requires and as “label count drives gain” does not.

**Robustness.** On CIFAR-100-C, corruption robustness tracks 55–60% of the clean gain at low data and is neutral at 100%; the prior neither buys nor costs robustness beyond its accuracy effect.

## S8. The prior’s gain across the data axis, drawn

The article gives these envelopes as numbers: the convolutional ones in its Tables 4 and 5, the ViT-tiny ones in its Table 6. This figure draws the same values and adds none of its own, because the shape of a curve is easier to see than to read off a grid.

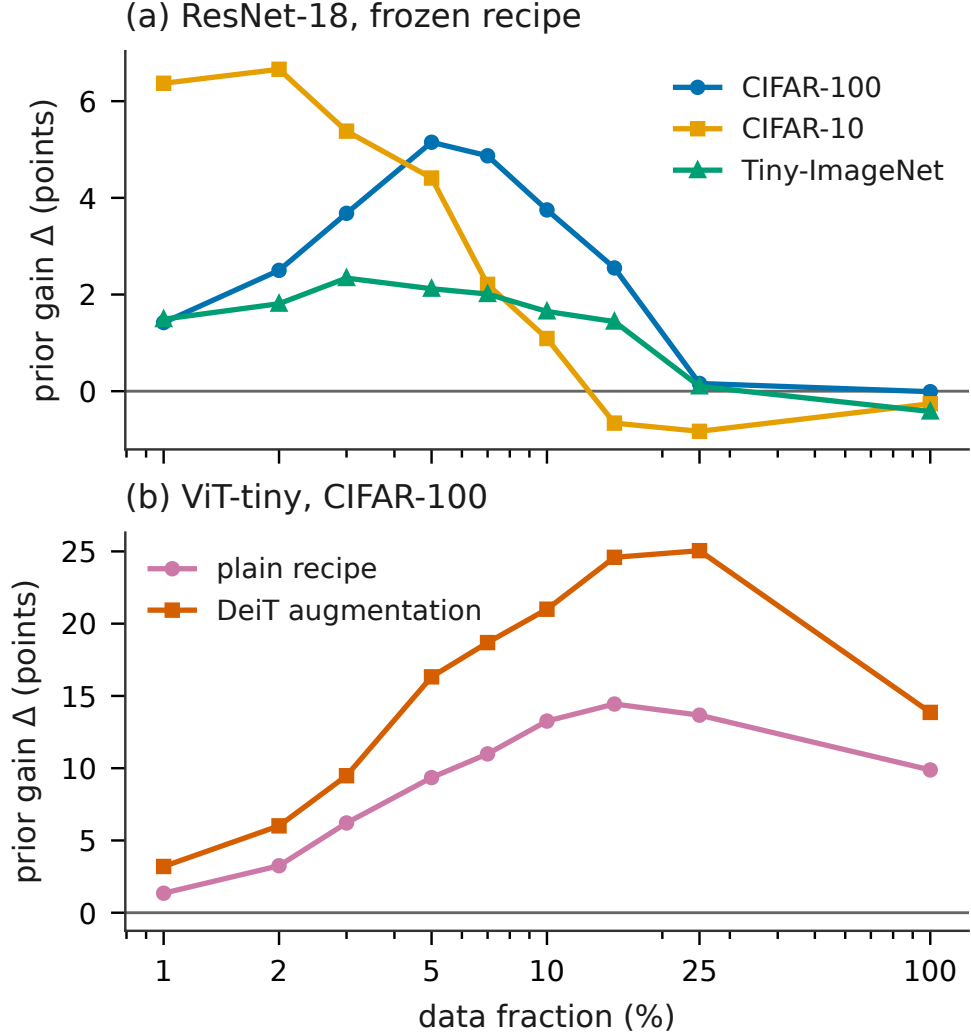


Figure 2: The prior’s gain across the data axis. **(a)** On convolutional backbones the envelope is unimodal and neutral at sufficiency under the frozen epoch-based recipe; the peak tracks task difficulty rather than fraction, and at a matched step budget the right flank does not appear at all (article, Sec. 3.8). **(b)** On ViT-tiny the gain is large at *every* scale, and DeiT-strength augmentation amplifies rather than replaces it.